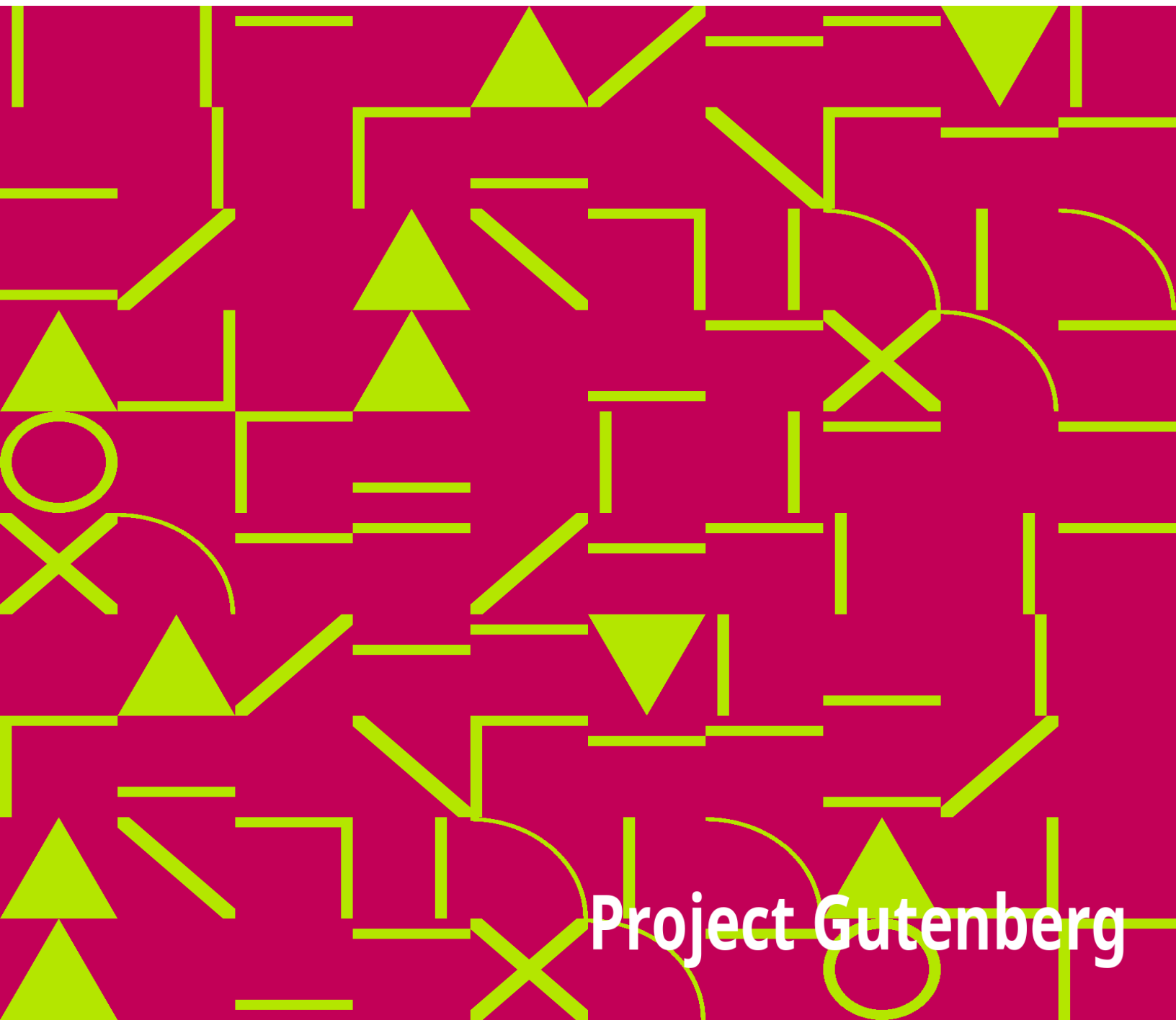


The Ancestry of Modern Amphibia: A Review of the Evidence

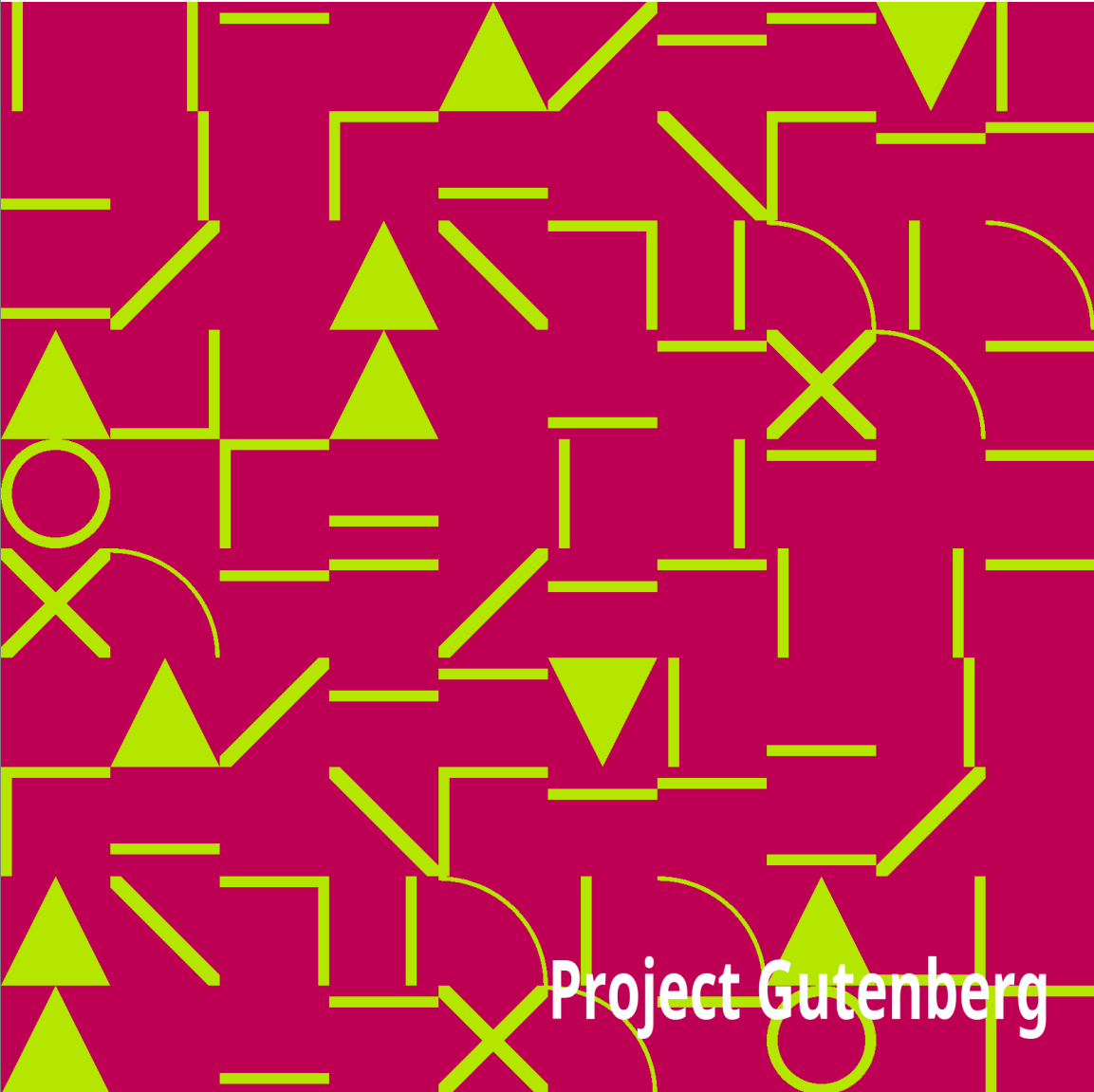
Theodore H. Eaton

A decorative background pattern consisting of various geometric shapes in black and white, including triangles, squares, circles, and lines, arranged in a complex, overlapping manner.

Project Gutenberg

The Ancestry of Modern Amphibia: A Review of the Evidence

Theodore H. Eaton

A decorative pattern consisting of various geometric shapes (triangles, squares, circles, and lines) in a light green color, scattered across a dark green background.

Project Gutenberg

The Project Gutenberg eBook of The Ancestry of Modern Amphibia: A Review of the Evidence

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

Title: The Ancestry of Modern Amphibia: A Review of the Evidence

Author: Theodore H. Eaton

Release date: September 8, 2011 [eBook #37350]

Language: English

Other information and formats: www.gutenberg.org/ebooks/37350

Credits: Produced by Chris Curnow, Charlene Taylor, Joseph Cooper
and the Online Distributed Proofreading Team at
<http://www.pgdp.net>

*** START OF THE PROJECT GUTENBERG EBOOK THE ANCESTRY OF MODERN
AMPHIBIA: A REVIEW OF THE EVIDENCE ***

**UNIVERSITY OF KANSAS PUBLICATIONS
MUSEUM OF NATURAL HISTORY**

Volume 12, No. 2, pp. 155-180, 10 figs.

-----July 10, 1959-----

**The Ancestry of Modern Amphibia:
A Review of the Evidence**

BY

THEODORE H. EATON, JR.

UNIVERSITY OF KANSAS
LAWRENCE
1959

UNIVERSITY OF KANSAS PUBLICATIONS, MUSEUM OF NATURAL HISTORY

Editors: E. Raymond Hall, Chairman, Henry S. Fitch, Robert W. Wilson

Volume 12, No. 2, pp. 155-180
Published July 10, 1959

UNIVERSITY OF KANSAS
Lawrence, Kansas

PRINTED IN THE STATE PRINTING PLANT
TOPEKA, KANSAS
1959



27-8362

[Transcriber's Notes: Several typos have been regulated.

One typo of "ancester" for "ancestor" was corrected.

One instance of "salamanderlike" corrected to "salamander-like".

The captions on some images give relative size. Due to differences in monitor size/resolution, do not consider the images to be scalable.]

The Ancestry of Modern Amphibia:

A Review of the Evidence

BY

THEODORE H. EATON, JR.

INTRODUCTION

In trying to determine the ancestral relationships of modern orders of Amphibia it is not possible to select satisfactory structural ancestors among a wealth of fossils, since very few of the known fossils could even be considered possible, and scarcely any are satisfactory, for such a selection. The nearest approach thus far to a solution of the problem in this manner has been made with reference to the Anura. Watson's paper (1940), with certain modifications made necessary by Gregory (1950), provides the paleontological evidence so far available on the origin of frogs. It shows that several features of the skeleton of frogs, such as the enlargement of the interpterygoid spaces and orbits, reduction of the more posterior dermal bones of the skull, and downward spread of the neural arches lateral to the notochord, were already apparent in the Pennsylvanian *Amphibamus* (Fig. 1), with which Gregory synonymized *Miobatrachus* and *Mazonerpeton*. But between the Pennsylvanian and the Triassic (the age of the earliest known frog, *Protobatrachus*) there was a great lapse of time, and that which passed between any conceivable Paleozoic ancestor of Urodela and the earliest satisfactory representative of this order (in the Cretaceous) was much longer still. The Apoda, so far as known, have no fossil record.

Nevertheless it should be possible, first, to survey those characters of modern Amphibia that might afford some comparison with the early fossils, and second, to discover among the known Paleozoic kinds those which are sufficiently unspecialized to permit derivation of the modern patterns. Further circumstantial evidence may be obtained by examining some features of Recent Amphibia which could not readily be compared with anything in the fossils; such are the embryonic development of the soft structures, including cartilaginous stages of the skeleton, the development and various specializations of the ear mechanism, adaptive characters associated with aquatic and terrestrial life, and so on.

COMPARISON OF MODERN ORDERS WITH THE LABYRINTHODONTS AND LEPOSPONDYLS

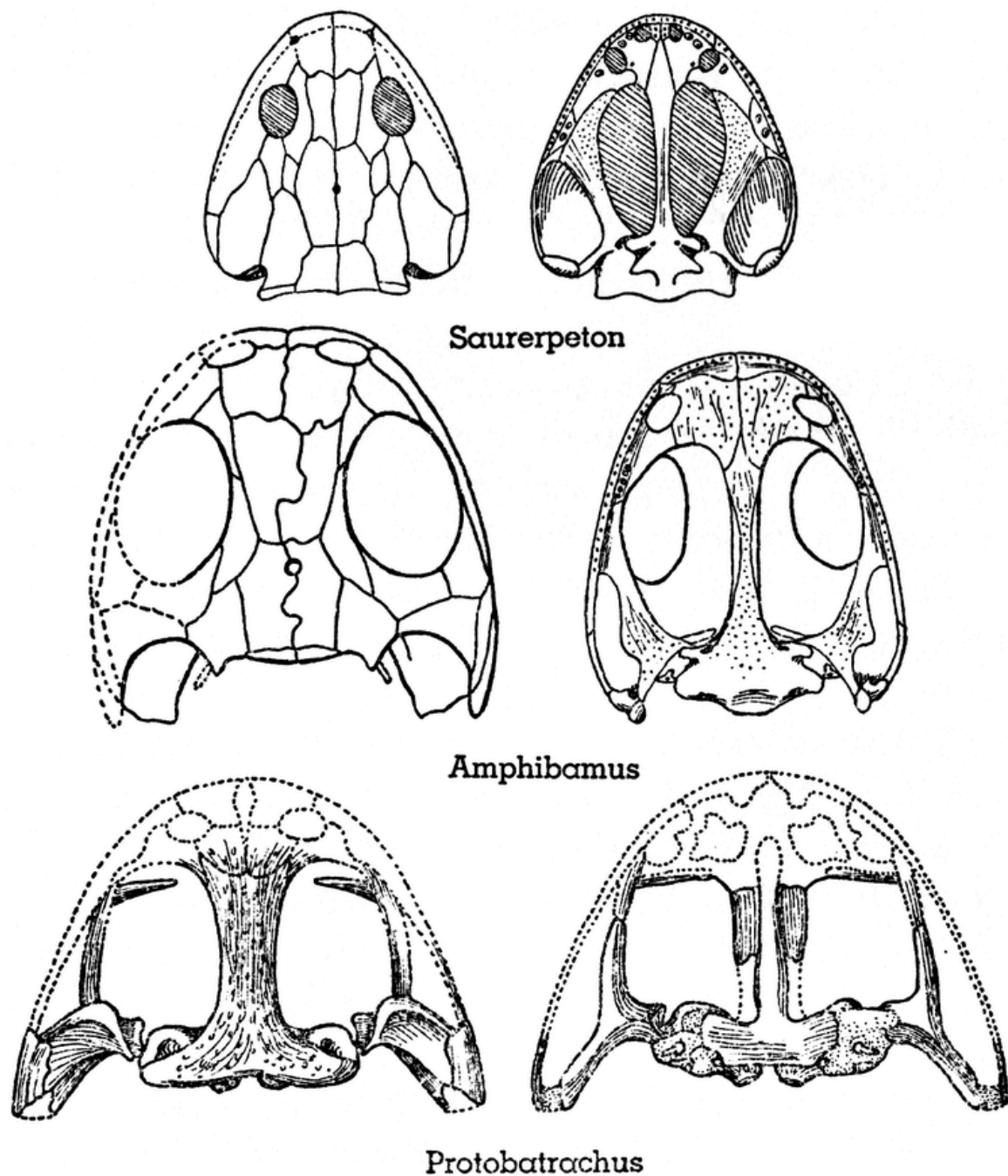


Fig. 1. Saurerpeton ($\times 1/2$, after Romer, 1930, fig. 6); Amphibamus, the palatal view $\times 2-1/4$, from Watson, 1940, fig. 4 (as Miobatrachus), the dorsal view $\times 2-1/2$, from Gregory's revised figure of Amphibamus (1950, Fig. 1); Protobatrachus, $\times 1$, from Watson, 1940, fig. 18, 19.

In both Anura and Urodela the skull is short, broad, relatively flat, with reduced pterygoids that diverge laterally from the parasphenoids leaving large interpterygoid vacuities, and with large orbits. (These statements do not apply to certain larval or perennibranchiate forms.) The skull in both orders has lost a number of primitive dermal bones in the posterior part; these are:

basioccipital, supraoccipital, postparietal, intertemporal, supratemporal, and tabular. The exoccipitals form the two condyles but there are no foramina for the 11th and 12th nerves, since these are not separate in modern Amphibia. The opisthotic is missing in all except Proteidae (but see discussion of the ear). Although the skull is normally autostylic, a movable basipterygoid articulation is present among Hynobiid salamanders and in at least the metamorphic stages of primitive frogs, and therefore should be expected in their ancestors. The vertebrae are, of course, complete; see discussion in later section. The quadratojugal, lost in salamanders, is retained in frogs, and conversely the lacrimal, absent in frogs, occurs in a few primitive salamanders. The situation in Apoda is different, but postfrontal and jugal should be noted as bones retained in this order while lost in the others.

Thus, in spite of minor differences, the above list shows that there are numerous and detailed similarities between Anura and Urodela with respect to the features in which they differ from the Paleozoic orders. Pusey (1943) listed 26 characters which *Ascaphus* shares with salamanders but not with more advanced frogs; a few of these might be coincidental, but most of them are of some complexity and must be taken to indicate relationship. The main adaptive specializations of Anura, however, including loss of the adult tail, extreme reduction in number of vertebrae, formation of urostyle, elongation of the ilium and lengthening of the hind legs, must have appeared at a later time than the separation of that order from any possible common stem with Urodela, although they are only partially developed in the Triassic *Protobatrachus*.

Turning to the Paleozoic Amphibia, there are two groups in which some likelihood of a relationship with modern order exists. In the Pennsylvanian Trimerorhachoidea (Labyrinthodontia, order Temnospondyli) some members, such as *Eugyrinus*, *Saurerpeton*, and notably *Amphibamus* (Fig. 1) had short, broad heads, an expansion of palatal and orbital openings, posterior widening of the parasphenoid associated with divergence of the pterygoids, a movable basipterygoid articulation, and reduction in size (but not loss) of the more posterior dermal bones of the skull. In recognition of Watson's (1940) evidence that these animals make quite suitable structural ancestors of frogs, Romer (1945) placed *Amphibamus* in an order, Eoanura, but Gregory (1950) indicated that it might better be left with the temnospondyls. Association of the urodele stem with this group does not seem to have been proposed hitherto.

The other group of Paleozoic Amphibia that has been considered probably ancestral to any modern type is the subclass Lepospondyli, containing three orders, Aistopoda, Nectridia and Microsauria. In these the vertebrae are complete (holospondylous), the centra presumably formed by cylindrical ossification around the notochord, and there is no evidence as to the contributions from embryonic cartilage units. It is important to note at this point that precisely the same statement can be made regarding the vertebrae of *adults* of all three Recent orders, yet for all of them, as shown in a later section, we have ample evidence of the part played by cartilage elements in vertebral development. Therefore (a) we cannot say that there were no such elements in embryonic stages of lepospondyls, and (b) it would take more than the evidence from adult vertebrae to relate a particular modern order (for example, Urodela) to the Lepospondyli. Vague similarities to Urodela have been noted by many authors in the Nectridia, Aistopoda and Microsauria, but these are not detailed and refer mainly to the vertebrae. The skulls do not show, either dorsally or in the palate, any striking resemblance to those of generalized salamanders, and certainly most known lepospondyls are too specialized to serve as the source of Urodela. It is true that the elongate bodies, small limbs, and apparent aquatic habitus of some lepospondyls accord well with our usual picture of a salamander, but such a form and way of life have appeared in many early Amphibia, including the labyrinthodonts. The family Lysorophidae (Fig. 2), usually placed among microsaurians, is

sufficiently close in skull structure to the Apoda to be a possible ancestor of these, but it probably has nothing to do with Urodela, by reason of the numerous morphological specializations that were associated with its snakelike habitus.

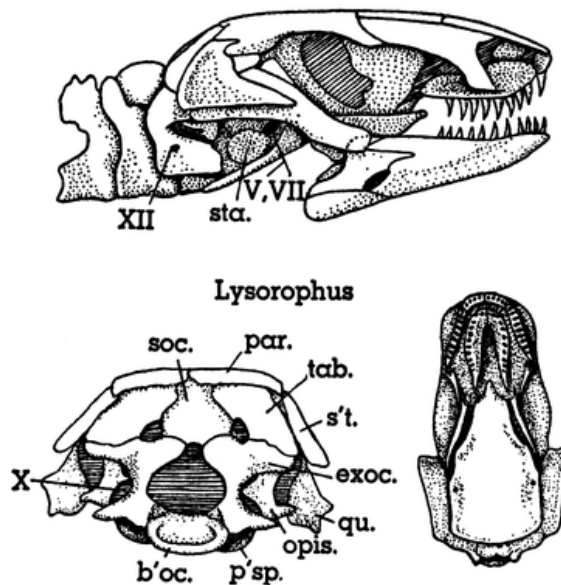


Fig. 2. *Lysorophus tricarinatus*, lateral and posterior views $\times 2\frac{1}{2}$, modified after Sollas, 1920, Figs. 8 and 12, respectively; palatal view after Broom, 1918, $\times 1\frac{1}{2}$. For explanation of abbreviations see Fig. 3.

the Urodela from *Porolepid Crossopterygii*, and all other tetrapods from the Rhipidistia; Säve-Söderbergh and Holmgren, the latter using the structure of carpus and tarsus, see a relationship of Urodela to Dipnoi, but accept the derivation of labyrinthodonts and other tetrapods from Rhipidistia. All of this work is most detailed and laborious, and has produced a great quantity of data useful to morphologists, but the diphyletic theory is not widely adopted; the evidence adduced for it seems to consist largely of minutiae which, taken by themselves, are inconclusive, or lend themselves to other interpretation. For instance Holmgren's numerous figures of embryonic limbs of salamanders show patterns of cartilage elements that he would trace to the Dipnoan type of fin, yet it is difficult to see that the weight of evidence requires this, when the pattern does not differ in any fundamental manner from those seen in other embryonic tetrapods, and the differences that do appear may well be taken to have ontogenetic rather than phylogenetic meaning. Further, the Dipnoan specialization of dental plates and autostylic jaw suspension, already accomplished early in the Devonian, would seem to exclude Dipnoi from possible ancestry of the Urodela, an order unknown prior to the Mesozoic, in which the teeth are essentially similar to those of late Paleozoic Amphibia, and the jaw suspension is not yet in all members autostylic.

McDowell's (1958) suggestion that it would be profitable to look among the Seymouriamorpha for the ancestors of frogs seems to be based upon a few details of apparent resemblance rather than a comprehensive view of the major characters of the animals. In most points which he mentions (limb girdles, form of ear, pterygoid articulation) the present writer does not see a closer similarity of frogs to Seymouriamorpha than to Temnospondyli.

Still other opinions have been expressed. Herre (1935), for instance, concludes "on anatomical, biological and paleontological grounds" that the orders of Urodela, Anura, Apoda and Stegocephali were all independently evolved from fish, but beyond citing the opinions of a number of other authors he does not present tangible evidence for this extreme polyphyletic interpretation.

More notable are the views of several Scandinavian workers (Säve-Söderbergh, 1934; Jarvik, 1942; Holmgren, 1933, 1939, 1949a, b), of whom Jarvik, in a thorough analysis of the ethmoid region, would derive

THE EAR

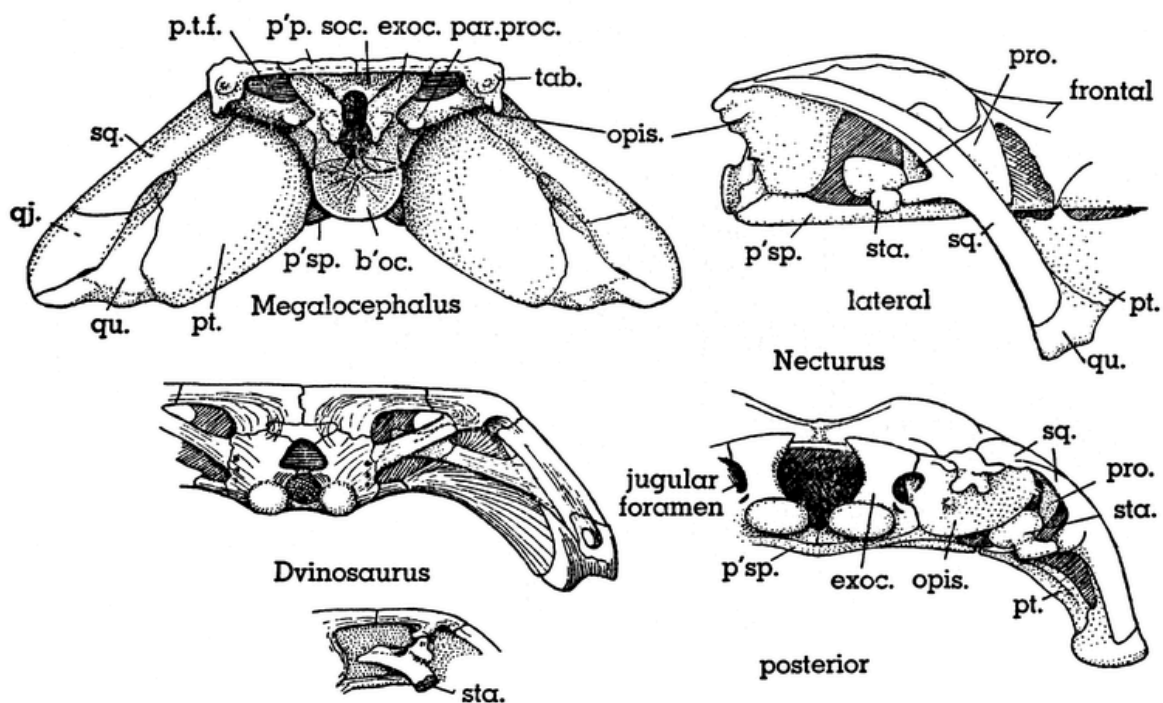


Fig. 3. Occipital region of skulls of *Megalocephalus brevicornis* ($\times 3/10$, after Watson, 1926, as *Orthosaurus*), *Dvinosaurus* ($\times 1/4$, modified after Bystrow, 1938; the lower figure after Sushkin, 1936), and *Necturus maculosus* ($\times 3$, original, from K. U., No. 3471).

Abbreviations Used in Figures

b'd.c.—basidorsal cartilage (neural)
 b'oc.—basioccipital
 ce.₁₋₄—centrale₁₋₄
 ch.—ceratohyal
 clav.—clavicle
 clei.—cleithrum
 cor.—coracoid
 d.c.₁₋₄—distal carpal₁₋₄
 diap.—diapophysis
 exoc.—exoccipital
 ep.—episternum
 hyost.—hyostapes
 i.—intermedium
 Mk.—Meckel's cartilage
 n.—notochord
 om.—omosternum
 op.—operculum
 opis.—opisthotic
 par.—parietal

par. proc.—paroccipital process
 peri. cent.—perichordal centrum
 p'p.—postparietal
 prep.—prepollex
 pro.—prootic
 p'sp.—parasphenoid
 pt.—pterygoid
 p.t.f.—post-temporal fossa
 postzyg.—postzygapophysis
 qj.—quadratojugal
 qu.—quadrate
 ra.—radiale
 r.hy.—hyomandibular ramus of VII
 rib-b.—rib-bearer
 r.md.—mandibular ramus of VII
 sc.—scapula
 sc'cor.—scapulocoracoid
 s'd.—supradorsal cartilage
 s'd.(postzyg.)—supradorsal (postzygapophysis)
 soc.—supraoccipital
 sp.c.—spinal cord
 sq.—squamosal
 s'sc.—suprascapula
 s't.—supratemporal
 sta.—stapes
 ster.—sternum
 tab.—tabular
 uln.—ulnare
 v.a.—vertebral artery
 xiph.—xiphisternum
 I, IV—digits I and IV
 V, VII, X, XII—foramina for cranial nerves of these numbers (in Fig. 4, VII is the facial nerve)

In temnospondylous Amphibia the tympanum generally occupied an otic notch, at a high level on the skull, bordered dorsomedially by the tabular and ventrolaterally by the squamosal. In this position the tympanum could receive airborne sounds whether the animal were entirely on land or lying nearly submerged with only the upper part of its head exposed. Among those Anura in which the ear is not reduced the same is true, except that the tabular is lost. In Temnospondyli (Fig. 3) the posterior wall of the otic capsule was usually formed by the opisthotic, which extended up and outward as a buttress from the exoccipital to the tabular, and sometimes showed a paroccipital process for the insertion, presumably, of a slip or tendon of the anterior axial musculature. The stapes, in addition to its foot in the fenestra ovalis and its tympanic or extrastapedial process to the tympanum, bore a dorsal process (or ligament) to the tabular, an "internal" process (or ligament) to the quadrate or an adjacent part of the squamosal, and a ligament to the ceratohyal. Some of these attachments might be reduced or absent in special cases, but they seem to have been the ones originally present both phylogenetically and embryonically in Amphibia.

Among typical frogs (Fig. 4) the base, or otostapes, is present and bony, the extrastapedial process (extracolumella, or hyostapes) is usually cartilaginous, the dorsal process (processus paroticus) is of cartilage or ligament, but the other two attachments are absent in the adult. The exoccipital extends laterally, occupying the posterior face of the otic capsule. Between it and the otostapes is a small disc, usually ossified, the operculum, which normally fits loosely in a portion of the fenestral membrane, and is developed from the otic capsule. The opercularis muscle extends from this disc to the suprascapula, in many but by no means all families of Anura.

Among Urodela (Fig. 4) the middle ear cavity and tympanum are lacking, and the stapes (columella) consists of no more than its footplate and the stylus, which is attached to the border of the squamosal, thus corresponding to the "internal" process. In families in which individuals metamorphose and become terrestrial (Hynobiidae, Ambystomidae, Salamandridae, Plethodontidae), an operculum and opercularis muscle appear in the adult, just as in frogs, except that in Plethodontidae, the most progressive family, the operculum fuses with the footplate of the stapes. Among neotenus or perennibranchiate urodeles there is no separate operculum or opercularis. The evidence given by Reed (1915) for fusion of the operculum with the columella in *Necturus* appears inconclusive, in spite of the great care with which his observations were made. On the other hand, *Necturus* and *Proteus* alone among living salamanders have a distinct opisthotic on the posterior wall of the otic capsule (Fig. 3), as do the Cretaceous *Hylaeobatrachus* and the Eocene *Palaeoproteus*. Probably these Proteidae should be regarded as primitive in this respect, although many other features may be attributed to neoteny.

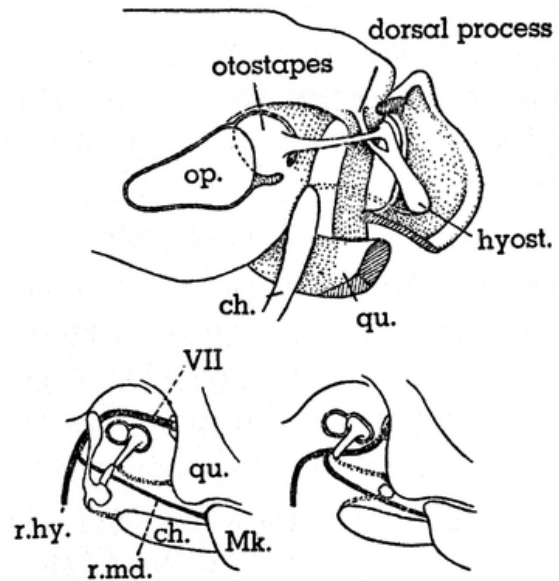


Fig. 4. Diagram of middle ear structures in *Rana* (upper figure, after Stadtmüller, 1936, and lower left after DeBeer, 1937), and *Ambystoma* (lower right, after DeBeer, 1937); all $\times 4$. For explanation of abbreviations see Fig. 3.

There is a contrast between Anura and most Urodela in the relative positions of the stapes and facial nerve, as shown in DeBeer's (1937) diagrams. In the latter (*Ambystoma*) the nerve is beneath, and in the former (*Rana*) above, the stapes. Judging by figures of *Neoceratodus*, *Hypogeophis*, and several types of reptiles and mammals, the Urodela are exceptional. *Necturus*, however, has the nerve passing above its stapes, and this may be primitive in the same sense as the persistent opisthotic. There can be, of course, no question of the nerve having worked its way through or over the obstructing stapes in order to come below it in salamanders; rather, the peripheral growth of neuron fibers in the embryo must simply pursue a slightly different course among the partially differentiated mesenchyme in the two contrasting patterns.

Although DeBeer (1937) shows in his figure of *Hypogeophis* (one of the Apoda) an operculum, this is apparently a mistake. The stapes has a large footplate, and its stylus articulates with the

quadrate, but no true operculum or opercularis has been described in the Apoda. The facial nerve passes above the stapes. It does not seem necessary to regard the conditions in this order as related directly to those of either salamanders or frogs, but a reduction of the stapes comparable to that in salamanders has occurred.

The presence in both frogs and terrestrial salamanders of a special mechanism involving the opercularis muscle and an operculum cut out in identical fashion from the wall of the otic capsule behind the stapes seems to require some other explanation than that of a chance convergence or parallelism. Although the stapes and otic region are readily visible in a number of labyrinthodonts and lepospondyls, no indication of an operculum seems to be reported among them. But in the Triassic *Protobatrachus* (Fig. 1), which is unmistakably a frog in its skull, pelvis and some other features, Piveteau (1937) has shown, immediately behind the foot of the stapes, a small bony tubercle, which he and Watson (1940) designated opisthotic. Very clearly it served for insertion of a muscle, and it is equally clear that the bone is a reduced opisthotic, carrying the paroccipital process already mentioned as characteristic of it in some temnospondyls. Since the remainder of the posterior wall of the otic capsule consists of cartilage, meeting the exoccipital, it may be that the opisthotic becomes the operculum in frogs. *Protobatrachus* was too far specialized in the Anuran direction, although it still had a tail, and the forelegs and hind legs were nearly the same size, to be considered a possible ancestor of the Urodeles. But at one stage in the general reduction of the skull in the ancestry of both groups, a condition similar to that in *Protobatrachus* may have characterized the otic region, long before the Triassic.

In the argument thus far we have considered terrestrial, adult amphibians, since it is only in these that either the normal middle ear and tympanum, or the opercular apparatus, is present. But among the urodeles several neotenic types occur (this term applies also to the perennibranchs). For most of these there is nothing about the otic region that would be inconsistent with derivation, by neoteny, from known families in which adults are terrestrial; for example, *Cryptobranchus* could have had a Hynobiid-like ancestor. But this, as mentioned above, does not hold for the Proteidae, which possess an opisthotic of relatively large size, distinctly separate from the exoccipital and prootic. Either this bone is a neomorph, which seems improbable, or there has not been in the ancestry of this particular family an episode of reduction comparable to that seen in the terrestrial families, where there is an operculum instead of a normal opisthotic. Therefore the Proteidae probably are not derived from the general stem of other salamanders, but diverged sufficiently long ago that the bones of the otic region were reduced on a different pattern. They need not be removed from the order, but, in this respect, recognized as more primitive than any other existing Urodela or Anura. A recent paper by Hecht (1957) discusses many features of *Necturus* and *Proteus*, and shows that they are remote from each other; his evidence does not seem to prove, however, that they were of independent origin or that they need be placed in separate families.

VERTEBRAE AND RIBS

Development of the vertebrae and ribs of Recent Amphibia has been studied by Gamble (1922), Naef (1929), Mookerjee (1930 a, b), Gray (1930) and Emelianov (1936), among others. MacBride (1932) and Remane (1938) provide good summaries. In this section reference will be made to the embryonic vertebral cartilages by the names used for them in these studies, although the concept of "arcualia" is currently considered of little value in comparative anatomy.

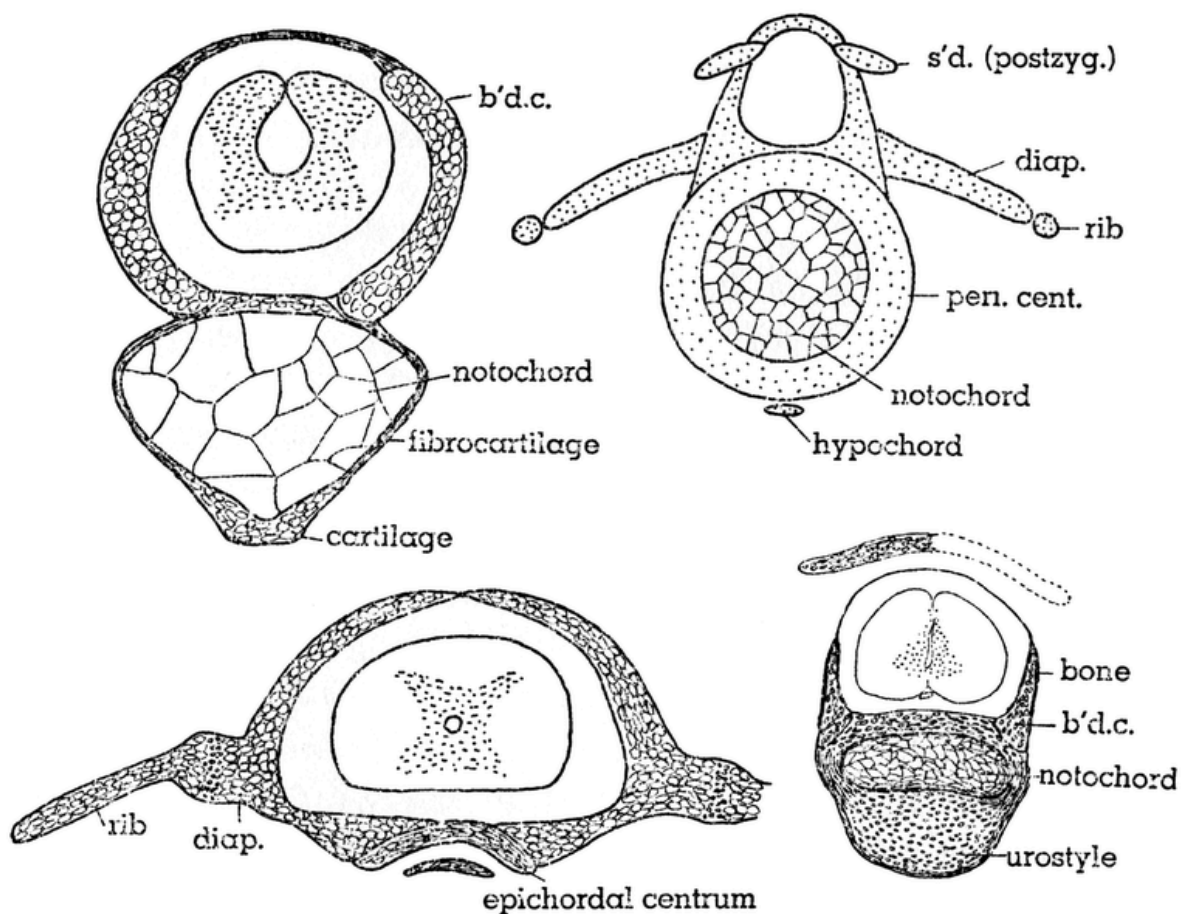


Fig. 5. Development of Anuran vertebrae. Upper left, late tadpole of *Xenopus laevis*; lower left, same just after metamorphosis; upper right, diagram of general components of primitive Anuran vertebra. (After MacBride, 1932, Figs. 35, 38, 47D, respectively.) Lower right, section through anterior portion of urostyle, immediately posterior to sacral vertebra, in transforming *Ascaphus truei* (original, from specimen collected on Olympic Peninsula, Washington). All $\times 20$ approx. For explanation of abbreviations see Fig. 3.

The centrum in Anura (Fig. 5) is formed in the perichordal sheath (*Rana*, *Bufo*) or only in the dorsal portion thereof (*Bombinator*, *Xenopus*). The neural arch develops from the basidorsal cartilages that rest upon, and at first are entirely distinct from, the perichordal sheath. Ribs, present as separate cartilages associated with the 2nd, 3rd and 4th vertebrae in the larvae of *Xenopus* and *Bombinator*, fuse with lateral processes (diapophyses) of the neural arches at metamorphosis, but in *Leiopelma* and *Ascaphus* the ribs remain freely articulated in the adult. Basiventral arcualia have been supposed to be represented by the hypochord, a median rod of cartilage beneath the shrinking notochord in the postsacral region, which at metamorphosis ossifies to produce the bulk of the urostyle. Fig. 5, lower right, a transverse section taken immediately posterior to the sacral ribs in a transforming specimen of *Ascaphus*, shows that the "hypochord" is a mass of cartilage formed in the perichordal sheath itself, and very obviously is derived from the ventral part of postsacral perichordal centra; there are, then, no basiventral arcualia, and the discrete hypochord shown in MacBride's diagram (Fig. 5, upper right) of a frog vertebra does not actually occur below the centrum, but only below the notochord in the postsacral region.

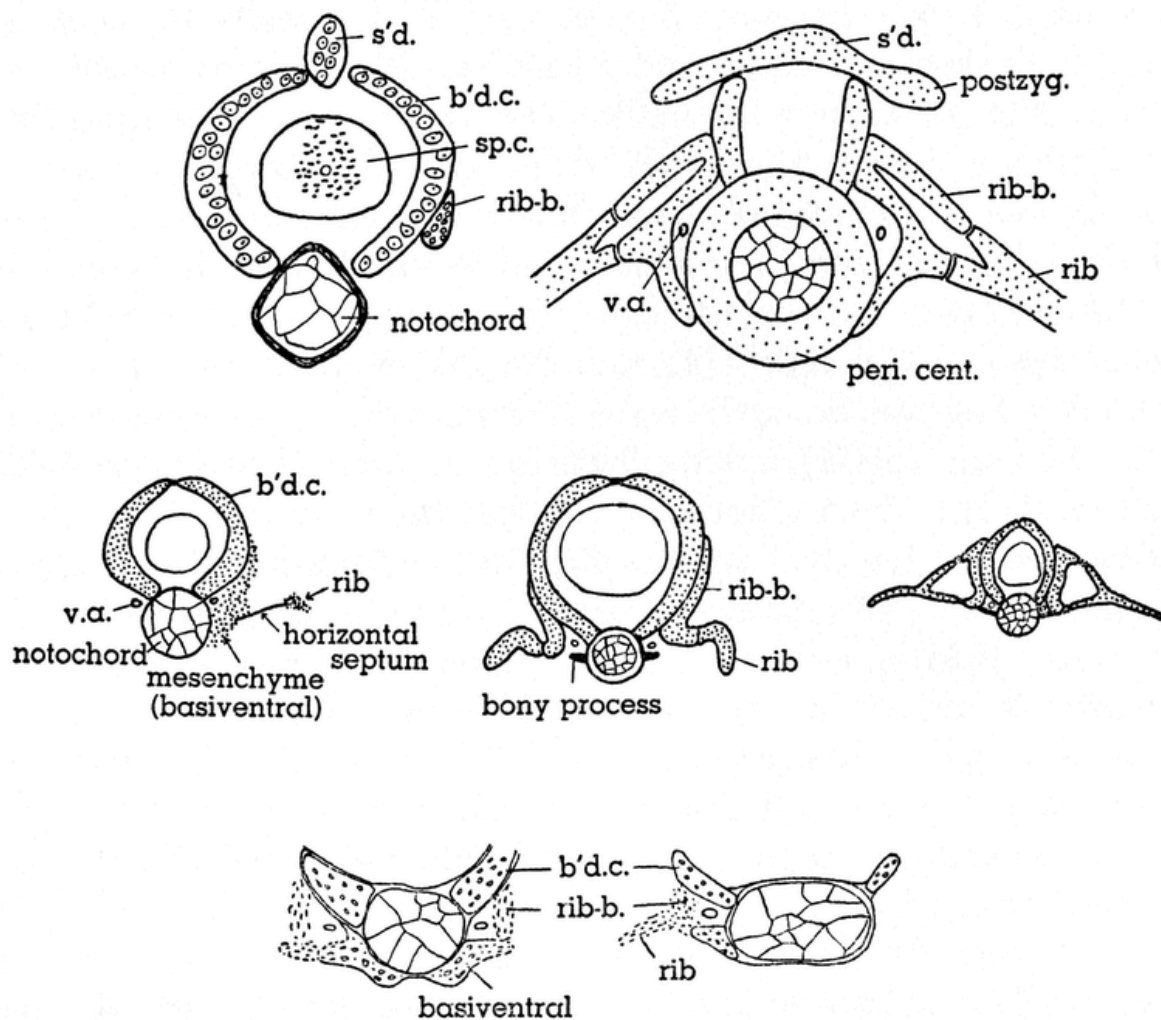


Fig. 6. Development of Urodele vertebrae. Upper figures, *Triton*: at left, larva at 20 mm., at right, diagram of components of vertebra (from MacBride, 1932, figs. 17, 47C). Middle figures, *Molge vulgaris* larva: left, at 18 mm.; middle, at 20-22 mm.; right, at 25 mm. (from Emelianov, 1936, figs. 33, 36, 38 respectively). Lower figures, *Necturus maculosus* larva: left, at 21 mm.; right, at 20 mm. (from MacBride, 1932, figs. 41.5, 41.3 respectively, after Gamble, 1922). All $\times 20$ approx. For explanation of abbreviations see Fig. 3.

In Urodela (Fig. 6) the pattern of vertebral and rib development is more complex, and there has been much controversy over its interpretation. Neural arches and perichordal centra form in the same manner as in frogs, but with the addition in certain cases (*Triton*) of a median supradorsal cartilage, which gives rise to the zygapophyses of each neural arch. Difficulty comes, however, in understanding the relationship of the ribs to the vertebrae. Each rib, usually two-headed, articulates with a "transverse process" that in its early development seems to be separate from both the vertebra and the rib, and is therefore known, noncommittally, as "rib-bearer." This lies laterally from the centrum, neural arch, and vertebral artery; upon fusing with the vertebra it therefore encloses the artery in a foramen separate from the one between the capitulum and tuberculum of the rib (the usual location of the vertebral artery). At least four different interpretations of these structures have been suggested:

(1) Naef (1929) considered the rib-bearer a derivative of the basiventral, which, by spreading laterally and dorsally to meet the neural arch, enclosed the vertebral artery. He then supposed that by reduction of the rib-bearer in other tetrapods (frogs and amniotes) the vertebralarterial foramen and costal foramen were brought together in a single foramen transversarium. The implication is that the Urodele condition is primitive, but it cannot now be supposed that Urodela are ancestral to any other group, and the rib-bearer is most probably a specialization limited to salamanders. This does not, of course, invalidate the first part of his interpretation.

(2) Remane (1938), noting that rib insertions of early Amphibia are essentially as in Amniota, argued that the rib-bearer is not from the basiventral but is a neomorph which originates directly from the neural arch and grows ventrally. This he inferred mainly from Gamble's (1922) observation on *Necturus*, but his assumption that *Necturus* is more primitive than other salamanders (such as the Salamandridae), where the pattern differs from this, is not necessarily correct. Rather, the perennibranchs are distinguished mainly by their neotenuous features, and their development is likely to show simplifications which are not necessarily primitive. The suggestion of a "neomorph" ought not to be made except as a last resort, for it is simply an acknowledgment that the author does not recognize homology with any structure already known; sometimes further information will make such recognition possible.

(3) Gray (1930), using *Molge taeniatus*, concluded that the normal capitulum of the rib was lost, but that the tuberculum bifurcated to make the two heads seen in Urodela, thus accounting for the failure of the costal foramen to coincide with that of the vertebral artery. This answer, too, seems to entail an unprovable assumption which should not be made without explicit evidence.

(4) Finally, Emelianov (1936) regarded the rib-bearer as a rudimentary *ventral* rib, on account of its relationship to the vertebral artery, and considered the actual rib to be a neomorph in the *dorsal* position characteristic of tetrapod ribs in general. This argument would fit the ontogenetic picture satisfactorily, provided that (a) there were some evidence of ventral, rather than dorsal, ribs in early Amphibia, and (b) we accept the invention of another neomorph in modern Amphibia as an unavoidable necessity. Emelianov's conclusion (p. 258) should be quoted here (translation): "The ribs of Urodela are shown to be upper ribs, yet we find besides these in Urodela rudimentary lower ribs fused with the vertebral column. The ribs of Apoda are lower ribs. In Anura ribs fail to develop fully, but as rare exceptions rudiments of upper ribs appear."

Of these various interpretations, that of Naef seems to involve the minimum of novelty, namely, that the rib-bearer is the basiventral, expanded and external to the vertebral artery. It is not necessary to take this modification as the ancestral condition in tetrapods, of course. The basiventral (=intercentrum) would merely have expanded sufficiently to provide a diapophysis for the tuberculum as well as the (primitive) facet for the capitulum. No neomorph appears under this hypothesis, which has the distinct advantage of simplicity.

Figures of early stages in vertebral development by the authors mentioned show that the basidorsals chondrify first, as neural arches, while a separate mass of mesenchyme lies externally and ventrally from these. This mesenchyme may chondrify either in one piece (on each side) or in two; in *Molge* the part adjacent to the centrum is ossified in the 20-mm. larva, and subsequently unites with the more dorsal and lateral cartilaginous part, while the rib, appearing farther out, grows inward to meet this composite "rib-bearer." In *Necturus* the mesenchyme below the neural arch differentiates into a cartilage below the vertebral artery (position proper to a basiventral), a bridge between this and the neural arch, and a rib, the latter two chondrifying later than the

"basiventral" proper. In the "axolotl" (presumably *Ambystoma tigrinum*) the rib-bearer grows downward from its first center of chondrification at the side of the neural arch (Emelianov, 1936).

Thus it appears that the simplest hypothesis to account for the rib-bearer is that (a) it is the basiventral, (b) it is recognizable just before chondrification as a mass of mesenchyme in contact with both the notochordal sheath and the basidorsal cartilage, (c) it may chondrify or ossify first in its ventral portion or in its dorsal portion, the two then joining before it fuses with the rest of the vertebra, (d) the enclosure of the vertebral artery is a consequence of the extension of the basiventral beyond the position occupied by it in primitive Amphibia, and (e) there is no indication that this took place in other orders than the Urodela.

It seems that the vertebrae in Urodela have at least the following components: perichordal centra, separate basidorsal cartilages, and basiventrals, which are somewhat specialized in their manner of development. The vertebrae of Anura develop in the fashion just described except that basiventrals are lacking. It would seem no more difficult to accept the derivation of salamander vertebrae from the temnospondylous type than it is in the case of frogs, if other evidence points to such an ancestry.

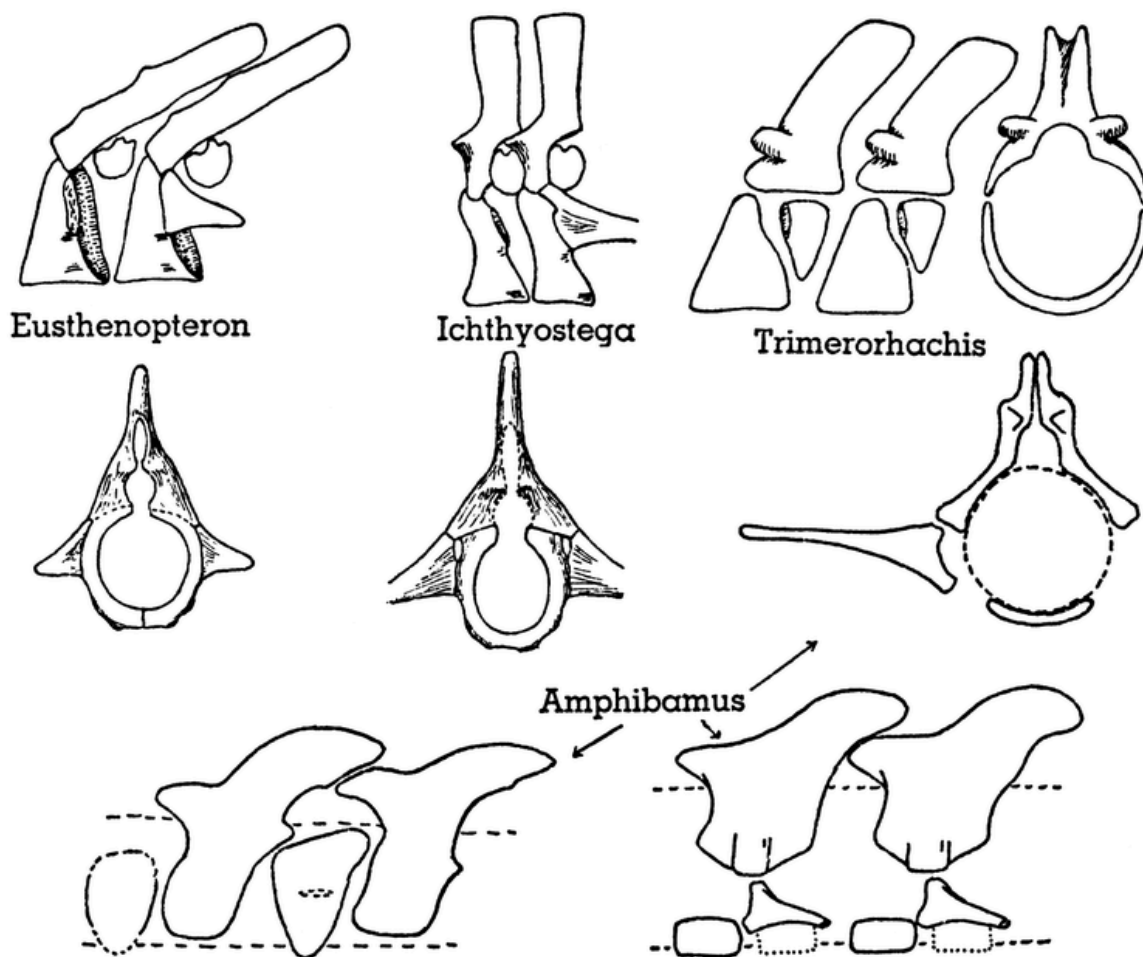


Fig. 7. Vertebrae of Eusthenopteron ($\times 1$) and Ichthyostega ($\times 2/3$, after Jarvik, 1952), Trimerorhachis ($\times 1-1/2$, after Case), and Amphibamus ($\times 10$, after Watson, 1940) in lateral

and end views; the two lower right-hand figures are from Watson (1940, as *Miobatrachus*); the lower left is from a cast of the "*Miobatrachus*" specimen in Chicago Natural History Museum, No. 2000, in the presacral region (original, $\times 10$).

Fig. 7, lower right, is Watson's (1940) illustration of the anterior trunk vertebrae of *Amphibamus* (*Miobatrachus*), in which the intercentrum is shown as a single median piece. Fig. 7, lower left, shows two of the more posterior trunk vertebrae seen as impressions in a cast of the type of "*Miobatrachus romeri*;" evidently the inter-centra were paired at about the level of the 16th vertebra, and relatively large. Gregory's (1950) figure of the type specimen of "*Mazonerpeton*" (also equivalent to *Amphibamus*) shows the anterior trunk vertebrae in relation to the ribs essentially as they appear to me in the cast of *Miobatrachus*, and rather differently from Watson's figure of the latter. Gregory is probably right in considering the specimens to represent various degrees of immaturity. So far as present information goes, then, the vertebrae of salamanders and frogs show no *clear* evidence of derivation from those of any particular group among the early Amphibia, but their features are not inconsistent with a simplification of the pattern of Temnospondyli.

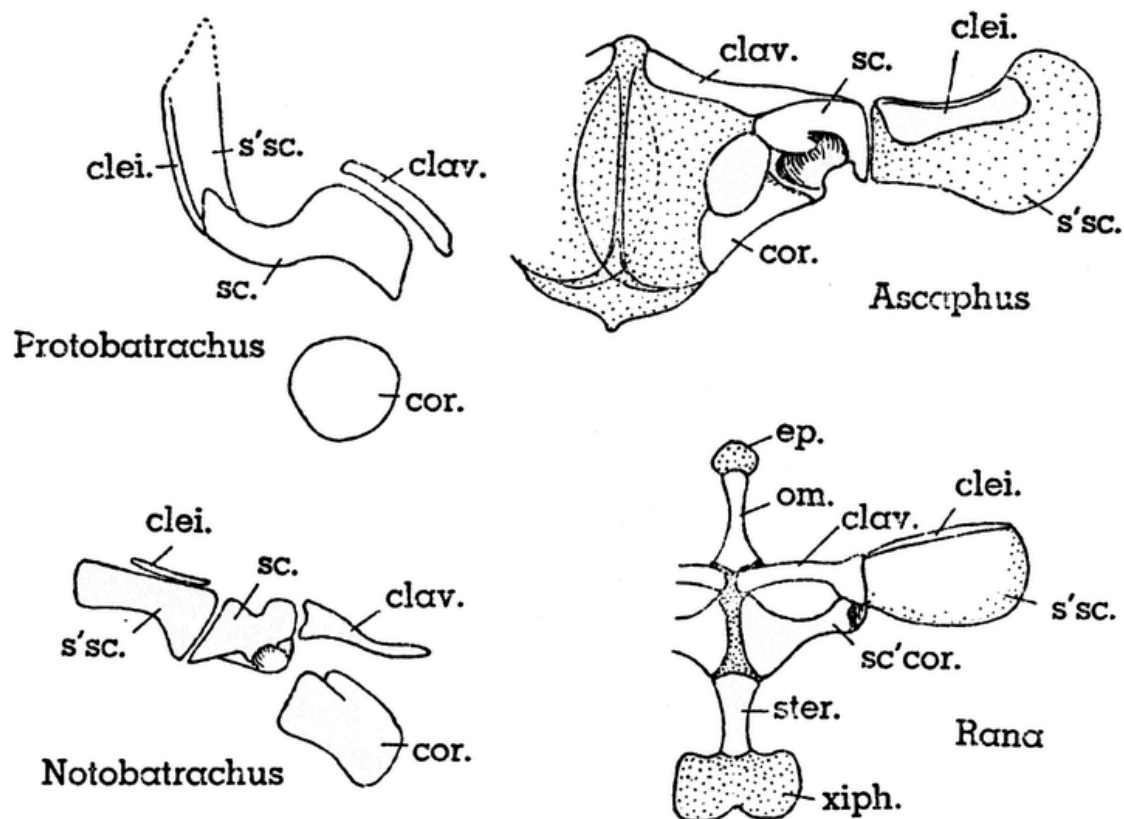


Fig. 8. Pectoral girdles of *Protobatrachus* (after Piveteau, 1937), *Notobatrachus* (after Stipanivic and Reig, 1956), *Ascaphus* (after Ritland, 1955 a) and *Rana* (original); all $\times 2$. For explanation of abbreviations see Fig. 3.

PECTORAL GIRDLE

Hecht and Ruibal (Copeia, 1928:242) make a strong point of the nature of the pectoral girdle in *Notobatrachus*, as described recently by Stipanovic and Reig (1955, 1956) from the Jurassic of Patagonia, and quite rightly recommend that the significance of the arciferal and firmisternal types of girdle be restudied. That of *Notobatrachus* is said to be firmisternal; in view of the arciferal condition in the supposedly primitive *Leiopelma*, *Ascaphus*, *Bombinator*, etc., this comes as a surprise. Is the firmisternal girdle, as seen in *Rana*, *Bufo*, and others, actually the ancestral type, and has the arciferal been derived from something like this?

In the figures given by Stipanovic and Reig the ossified parts of the girdle are figured in detail (Fig. 8) and Reig's discussion of it is thorough. The decision to call it firmisternal was taken with some hesitancy, for no median elements are indicated, and the position and shape of those seen is closely similar to the ossified parts in *Ascaphus* and *Leiopelma*; there is no bony sternum or omosternum. It is safe to suppose that some cartilage lay in the midline between the clavicles and coracoids, but there is no evidence as to its extent, rigidity, or degree of overlapping if any. Apparently, then, there is not sufficient reason to infer that this Jurassic frog had a pectoral girdle comparable with the modern firmisternal type.

Piveteau (1955:261) remarks that the only living Anuran that can be compared usefully with *Protobatrachus* (Triassic) with regard to its pectoral girdle is *Ascaphus*. Again, the extent of cartilage in *Protobatrachus* (Fig. 8) can only be inferred, and there are no median elements. The agreement with *Ascaphus* includes the presence, in both, of a separate coracoid ossification situated posterior to the ossified "scapulocoracoid" (actually scapula). This ossification is evidently that shown in *Notobatrachus* as "coracoid." Direct comparison of the three genera with one another suggests that if we use the term arciferal for any, we should use it for all.

In the remote predecessor of Anura, *Amphibamus* of the Pennsylvanian, the pectoral girdle was less substantial than in many of its contemporaries, but it contained the primitive median interclavicle in addition to the clavicle, cleithrum, and scapulocoracoid. (The figure of Watson, 1940, and that by Gregory, 1950, are of individuals of different ages, the latter being older.) It is clear that the paired elements of such a girdle were held rigid by their attachment to the interclavicle, *via* the clavicles. Subsequent elimination of the interclavicle in the Anuran line of descent, and decrease of ossification, left a girdle like that of *Protobatrachus*, *Notobatrachus*, *Ascaphus* and *Leiopelma*. But in several advanced families a more rigid median "sternum," of one or two bony pieces plus cartilage, is developed secondarily, possibly (as Cope, 1889: 247, suggested) in correlation with axillary amplexus.

Among Urodela no dermal bones occur in the pectoral girdle. There is usually a scapulocoracoid ossified as a single piece, from which a thin cartilaginous suprascapula extends dorsally and a broad cartilaginous coracoid plate extends medially, overlapping the one from the opposite side; a precoracoid lobe of this reaches forward on either side, and a median, posterior "sternum" of cartilage may make contact with the edges of the two coracoids. In *Siren* and *Amphiuma* two centers of ossification are found for each scapulocoracoid, and in *Triton* and *Salamandra* three. Probably the more dorsal and lateral of these represents the primitive scapula and the other one (or two) the primitive coracoid.

Comparing the girdle of a salamander with that of a frog, the closest similarity can be seen between *Ascaphus* and a salamander in which the scapula and coracoid ossify separately. Both

have the median "sternum" in contact with the coracoid plates. The major difference, of course, is the lack of clavicle and cleithrum in the salamander.

CARPUS AND TARSUS

In *Ascaphus* (Ritland, 1955a; cleared and stained specimens of nearly grown males) distal carpals 1, 2, 3 and 4 are present and separate, increasing in size in the order given (Fig. 9). A prepollex rests against centrale 1; centralia 2 and 3 are fused; the radiale fuses with centrale 4, and the intermedium fuses with the ulnare; radius and ulna are fused with each other as in other frogs. The digits (and metacarpals) are considered by Ritland to be 1-4, in addition to the prepollex, rather than 2-5.

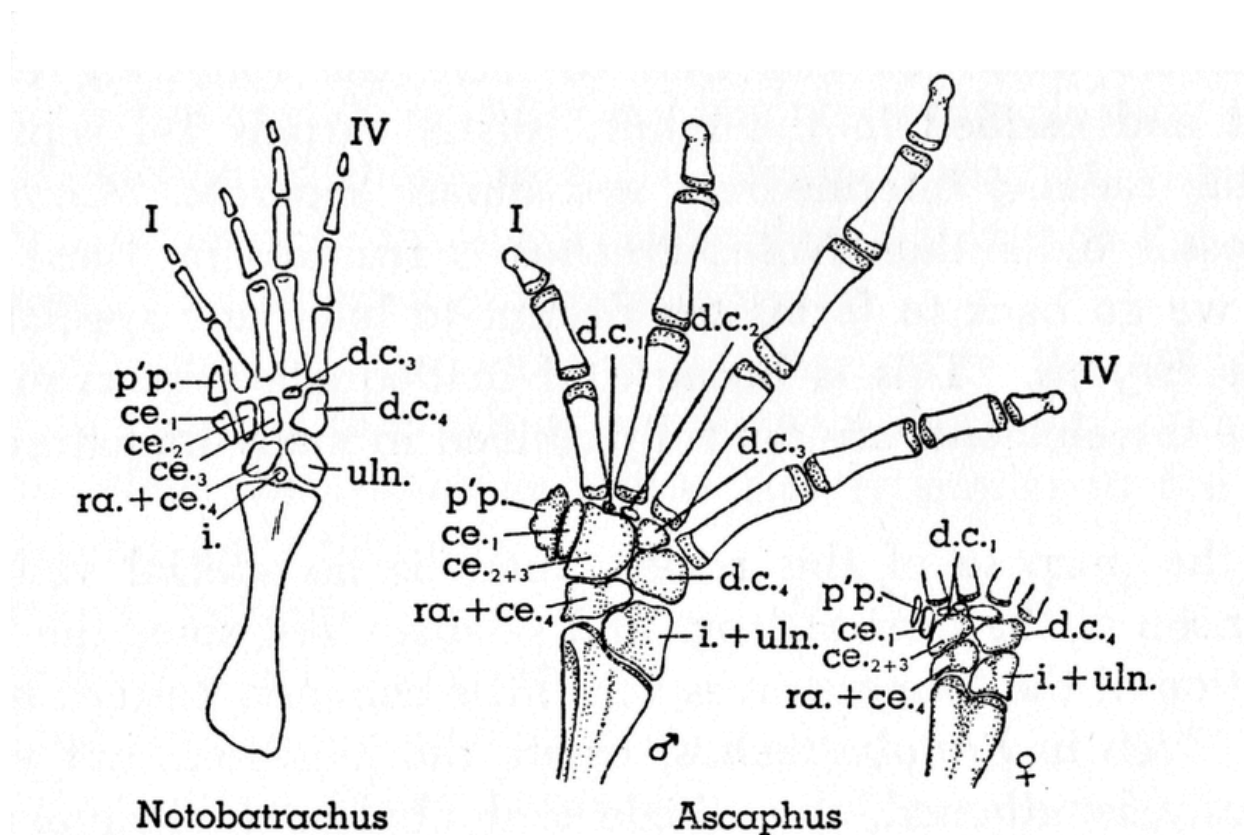


Fig. 9. Skeleton of fore foot of *Notobatrachus* (after Stipanovic and Reig, 1956, terminology revised) and *Ascaphus* (after Ritland, 1955 a); all $\times 5$. For explanation of abbreviations see Fig. 3.

In the Jurassic *Notobatrachus* Stipanovic and Reig (1956) have shown the carpus with surprising clarity (Fig. 9). If their nomenclature of the parts be revised, we obtain a fairly close resemblance to *Ascaphus*, except that centralia 2 and 3 are not fused, distal carpals 1 and 2 do not show (which would easily be understood if they were of the size of those in *Ascaphus*, or not ossified), and the intermedium remains separate from the ulnare.

In *Salamandra* (Francis, 1934; Nauck, 1938) distal carpals 1 and 2 are fused in both larva and adult, and 3 and 4 are separate; the radiale, intermedium and ulnare are separate in the larva but the latter two fuse in the adult; centrale 1 (labelled prepollical cartilage by Francis) and centrale 2 are separate. Francis considers the digits (and metacarpals) to be 1-4. Apparently the arrangement here indicated for the larva is characteristic of other larval salamanders, except where further reduced, and reduction below the number given for the adult is common in other terrestrial forms. The radius and ulna are, of course, separate.

The ossification of carpals is more likely to be complete in adult frogs than in salamanders, but some ossification of all parts named is found in several of the latter. A common ancestor of frogs and salamanders could be expected to have the following elements present and ossified in the adult: distal carpals 1-4 separate; 3 centralia; radiale, intermedium and ulnare separate. Comparison with fossils older than *Notobatrachus* is fruitless on these points, unless we go back to forms too distant to have any special value, such as *Eryops*. This is because of inadequate preservation and because the elements are not fully ossified in many immature specimens.

For the purpose of this review there is no special value in a comparison of the tarsi of frogs and salamanders, since the leaping adaptation of the former leaves very little common pattern between them. Even in *Protobatrachus*, where the legs were not yet conspicuously lengthened, the tibiale and fibulare ("astragalus" and "calcaneum" respectively) were already considerably elongated. The carpus and tarsus of *Amphibamus* are as yet undecipherable.

THE LARVA

Considering the postembryonic developmental stages of modern Amphibia, there can be no doubt that a gill-bearing, four-legged larva of a salamander, in which lateral line pores and a gular fold are present, represents much more closely the type of larva found in labyrinthodonts than does the limbless, plant-nibbling tadpole of the Anura. Salamander-like larvae of labyrinthodonts are well known, especially those formerly supposed to comprise the order Branchiosauria. Many, perhaps the majority of, labyrinthodonts show some features associated with aquatic life even when full-grown, as do the lepospondyls. These features may include impressions of sensory canals on the dermal bones of the skull, persistence of visceral arches, reduction in size of appendages, and failure of tarsal and carpal elements to ossify. In fact, it appears that very few of the Paleozoic Amphibia were successful in establishing themselves as terrestrial animals even as adults.

Nevertheless, in the ancestry of Anura, and that of at least the Hynobiid, Ambystomid, Salamandrid and Plethodontid salamanders, there must certainly have been a terrestrial adult, transforming from an aquatic larva. The leaping mechanism of Anura, shown in so many features of their anatomy, is perhaps to be explained as a device for sudden escape from land into the water, but it was not yet perfected in the Triassic *Protobatrachus* or the Jurassic *Notobatrachus*.

The middle ear, its sound-transmitting mechanism, and the tympanum, well developed in most Anura, are readily derived from those of early labyrinthodonts, and are presumably effective for hearing airborne sounds whether on land or while floating in the water. Reduction of these organs in Urodela may be correlated with their customary restriction to subsurface habitats and inability to maintain a floating position while in water.

Some light may be shed on the significance of the tadpole of Anura by considering the early stages of the ribbed frogs, Liopelmidae. *Leiopelma* and *Ascaphus* are so closely similar in the adult that there is no doubt that they belong in one family, primitive in some respects (including articulated ribs; pyriformis and caudalipuboischiotibialis muscles) but not in others (absence of tympanum and middle ear). In both genera the eggs are large, 5 mm. in *Leiopelma*, 4.5 mm. in *Ascaphus*, and unpigmented; but at this point the resemblance ends.

Stephenson (1955) showed that embryos of *L. hochstetteri* develop equally well on land (in damp places) or in the water, and that embryos prematurely released from egg capsules develop successfully in the water. The larvae possess both pairs of legs (Fig. 10) and a broad gular fold similar to that of larval salamanders. In *L. hochstetteri* the fold grows back over the forelegs temporarily, but remains free from the body and presently the legs reappear, whereas in *L. archeyi* the forelegs are not covered at any time. No branchial chamber or spiracle is formed. Of course direct development, without a tadpole, occurs in several other groups of Anura, but in each case terrestrial adaptations are obvious. This is not true of *Leiopelma*, which Stephenson regards as more nearly comparable with Urodela in its development than with other Anura, and he sees in it a "primary and amphibious" mode instead of a terrestrial specialization.

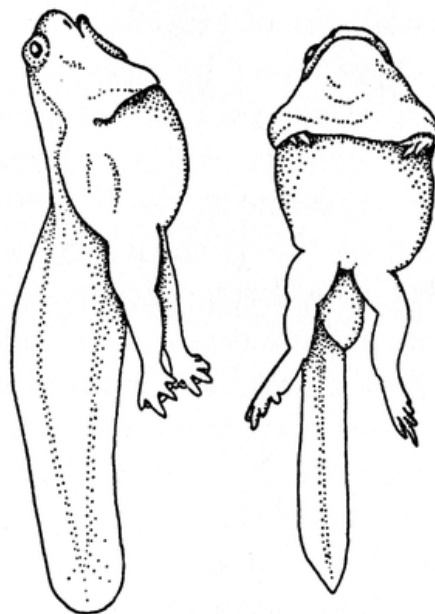


Fig. 10. *Leiopelma hochstetteri* larva, lateral and ventral (after Stephenson, 1955), $\times 4$.

The *Ascaphus* tadpole bears no outward resemblance to the larva of *Leiopelma*, but is a normal tadpole in form, although sluggish in activity. Its greatly expanded labial folds bear numerous rows of horny epidermal "teeth," which, with the lips, serve to anchor the tadpole to stones in the swift water of mountain brooks. Noble (1927) noticed that particles of food were taken in through the external nares, and that a current of water passed through these openings and out by way of the median spiracle. It appears that any action by the teeth and jaws in scraping algae from the rocks (which were bare in the stream where I have collected *Ascaphus*) would be quite incidental, and that the lips and teeth must be primarily a clinging mechanism. Certain other mountain brook tadpoles (for example, *Borborocoetes*) show similar devices, but these are developed independently, as specializations from the usual sort of tadpole.

May it not be that closure of the gill-chamber by the opercular (=gular) fold, retardation of limb development, expansion of the lips, growth of parallel rows of horny teeth, and other correlated features that make a tadpole, were brought about as an adaptation of the primitive Anuran larva to a swift-stream habitat, and that this "basic patent" then later served to admit the tadpoles of descendant types to an alga-scraping habit in quiet water as well? The tadpole, as a unique larval type among vertebrates, bears the hallmarks of an abrupt adaptive shift, such as might have occurred within the limits of a single family, and it seems difficult to imagine the enclosed branchial chamber, the tooth-rows, and lips of a familiar tadpole as having evolved without some kind of suctorial function along the way.

SUMMARY

The Anura probably originated among temnospondylous labyrinthodonts, through a line represented approximately by *Eugyrinus*, *Amphibamus*, and the Triassic frog *Protobatrachus*, as shown by Watson, Piveteau and others. The known Paleozoic lepospondyls do not show clear indications of a relationship with Urodela, but *Lysorophus* may well be on the ancestral stem of the Apoda.

Between Urodela and Anura there are numerous resemblances which seem to indicate direct relationship through a common stock: (1) a similar reduction of dermal bones of the skull and expansion of palatal vacuities; (2) movable basiptyergoid articulation in primitive members of both orders; (3) an operculum formed in the otic capsule, with opercularis muscle; (4) many details of cranial development, cranial muscles, and thigh muscles, especially between *Ascapheus* and the Urodela, as shown by Pusey and Noble; (5) essentially similar manner of vertebral development, quite consistent with derivation of both orders from Temnospondyli; (6) presence in the larva of *Leiopelma* of a salamander-like gular fold, four limbs, and no suggestion of modification from a tadpole (Stephenson).

LITERATURE CITED

BROOM, R.

1918. Observations on the genus *Lysorophus* Cope. Ann. Mag. Nat. Hist., (9)2:232-239.

BYSTROW, A. P.

1938. Dvinosaurus als neotenische Form der Stegocephalen. Acta Zool., 19:209-295.

CASE, E. C.

1935. Description of a collection of associated skeletons of *Trimerorhachis*. Contrib. Mus. Pal. Univ. Michigan, 4:227-274.

COPE, E. D.

1889. The Batrachia of North America. Bull. U. S. Nat. Mus., 34:1-525.

DE BEER, G. R.

1937. The development of the vertebrate skull. Pp. xxiii + 552. Oxford, Clarendon Press.

DE VILLIERS, C. G. S.

1934. Studies of the cranial anatomy of *Ascapthus truei* Stejneger, the American "Liopelmid." Bull. Mus. Comp. Zool., 77:1-38.

EMELIANOV, S. W.

1936. Die Morphologie der Tetrapodenrippen. Zool. Jahrb. (Anat.), 62:173-274.

FRANCIS, E. T. B.

1934. The anatomy of the salamander. Pp. xxxi + 381. Oxford, Clarendon Press.

GAMBLE, D. L.

1922. The morphology of the ribs and the transverse processes of *Necturus maculatus*. Jour. Morph., 36:537-566.

GRAY, P.

1930. On the attachments of the Urodele rib to the vertebra and their homologies with the capitulum and tuberculum of the Amniote rib. Proc. Zool. Soc. London, 1930(1931):907-911.

GREGORY, J. T.

1950. Tetrapods from the Pennsylvanian nodules from Mazon Creek, Illinois. Am. Jour. Sci., 248:833-873.

HECHT, M. E.

1957. A case of parallel evolution in salamanders. Proc. Zool. Soc. Calcutta, Mookerjee Mem.:283-292.

HOLMGREN, N.

1933. On the origin of the tetrapod limb. Acta Zool., 14:185-295.

1939. Contributions to the question of the origin of the tetrapod limb. Acta Zool., 20:89-124.

1949a. Contributions to the question of the origin of tetrapods. Acta Zool., 30:459-484.

1949b. On the tetrapod limb problem again. Acta Zool., 30:485-508.

HERRE, W.

1935. Die Schwanzlurche der mitteleocänen (oberlutetischen) Braunkohle des Geiseltales und die Phylogenie der Urodelen unter Einschluss der fossilen Formen. Zoologica, 33:87, 1-5.

JARVIK, E.

1942. On the structure of the snout of Crossopterygians and lower gnathostomes in general. Zool. Bidrag fran Upsala, 21:235-675. 1952. On the fish-like tail in the Ichthyostegid Stegocephalians. Meddelelser on Grønland, 114(12):1-90.

MOOKERJEE, H.K.

1930a. On the development of the vertebral column of the Urodela. Phil. Trans. Roy. Soc. London, B 218:415-446.

1930b. On the development of the vertebral column of the Anura. Philos. Trans. Royal Soc. London, B 219:165-196.

MACBRIDE, E.W.

1932. Recent work on the development of the vertebral column. Cambridge, Biol. Rev., 7:108-148.

McDOWELL, S.B.

1958. Are the frogs specialized seymouriamorphs? (Abstract) Anat. Rec., 132(3):472.

NAEF, A.

1929. Notizen zur Morphologie und Stammesgeschichte der Wirbeltiere. 15. Dreissig Thesen über Wirbelsäule und Rippen, insbesondere bei den Tetrapoden. Zool. Jahrb., 50:581-600.

NOBLE, G. K.

1922. The phylogeny of the Salientia; I. The osteology and the thigh musculature; their bearing on classification and phylogeny. Bull. Amer. Mus. Nat. Hist., 46:1-87.

1927. The value of life-history data in the study of the evolution of the Amphibia. Annals New York Acad. Sci., 30:31-128.

PIVETEAU, J.

1937. Un Amphibien du Trias inférieur. Essai sur l'origine et l'évolution des Amphibiens Anoures. Annales de Paléontologie, 26:135-176.

1955. Anoura. In: Traité de Paléontologie, 5:250-274. J. Piveteau, Masson et Cie, Paris.

PUSEY, H. K.

1943. On the head of the liopelmid frog, *Ascaphus truei*. I. The chondrocranium, jaws, arches, and muscles of a partly grown larva. Quart. Jour. Micr. Sci., 84:105-185.

REED, H. D.

1915. The sound-transmitting apparatus in Necturus. Anat. Rec., 9:581-590.

REMANE, A.

1936. Wirbelsäule und ihre Abkömmlinge. In: Handbuch der vergleichenden Anatomie der Wirbeltiere, L. Bolk *et al.*, 4:1-206. Urban and Schwarzenberg, Berlin, Vienna.

RITLAND, R. M.

1955a. Studies on the post-cranial morphology of *Ascaphus truei*. I. Skeleton and spinal nerves. Jour. Morph., 97:119-174.

1955b. Studies on the post-cranial morphology of *Ascaphus truei*. II. Myology. Jour. Morph., 97:215-282.

ROMER, A. S.

1945. Vertebrate paleontology. 2nd edition. Pp. viii + 687. Univ. Chicago Press.

1947. Review of the Labyrinthodontia. Bull. Mus. Comp. Zool., 99:3-368.

SÄVE-SÖDERBERGH, G.

1934. Some points of view concerning the evolution of the vertebrates and the classification of this group. Arkiv för Zoologi, 26A:1-20.

STADTMÜLLER, F.

1936. Kraniaum und Visceralskelett der Stegocephalen und Amphibien. In: Handbuch der vergleichenden Anatomie der Wirbeltiere, by L. Bolk *et al.*, 4:501-698.

STEPHENSON, N. G.

1955. On the development of the frog, *Leiopelma hochstetteri* Fitzinger. Proc. Zool. Soc. London, 124(4):785-795.

STIPANICIC, P. N. and REIG, O. A.

1955. Breve noticia sobre el hallazgo de anuros en el denominado "Complejo Porfirico de la Patagonia Extraandina," con consideraciones acerca de la composicion geologica del mismo. Revista de la Asoc. Geol. Argentina, 10(4):215-233.

1956. El "complejo porfirico de la Patagonia extraandina" y su fauna de Anuros. Acta Geol. Lilloana (Univ. Nac. del Tucuman), 1:185-297.

SUSHKIN, P. P.

1936. Notes on the pro-Jurassic Tetrapoda from U. S. S. R. III. Dvinosaurus Amalitzky, a perennibranchiate stegocephalian from the Upper Permian of North Dvina. Trav. Inst. Pal. Acad. Sci. URSS, 5:43-91.

WATSON, D. M. S.

1940. The origin of frogs. Trans. Roy. Soc. Edinburgh, 40(7):195-231.

Transmitted April 7, 1959.

*** END OF THE PROJECT GUTENBERG EBOOK THE ANCESTRY
OF MODERN AMPHIBIA: A REVIEW OF THE EVIDENCE ***

Updated editions will replace the previous one—the old editions will be renamed.

Creating the works from print editions not protected by U.S. copyright law means that no one owns a United States copyright in these works, so the Foundation (and you!) can copy and distribute it in the United States without permission and without paying copyright royalties. Special rules, set forth in the General Terms of Use part of this license, apply to copying and distributing Project Gutenberg™ electronic works to protect the PROJECT GUTENBERG™ concept and trademark. Project Gutenberg is a registered trademark, and may not be used if you charge for an eBook, except by following the terms of the trademark license, including paying royalties for use of the Project Gutenberg trademark. If you do not charge anything for copies of this eBook, complying with the trademark license is very easy. You may use this eBook for nearly any purpose such as creation of derivative works, reports, performances and research. Project Gutenberg eBooks may be modified and printed and given away—you may do practically ANYTHING in the United States with eBooks not protected by U.S. copyright law. Redistribution is subject to the trademark license, especially commercial redistribution.

START: FULL LICENSE

THE FULL PROJECT GUTENBERG™ LICENSE

PLEASE READ THIS BEFORE YOU DISTRIBUTE OR USE THIS WORK

To protect the Project Gutenberg™ mission of promoting the free distribution of electronic works, by using or distributing this work (or any other work associated in any way with the phrase “Project Gutenberg”), you agree to comply with all the terms of the Full Project Gutenberg License available with this file or online at www.gutenberg.org/license.

Section 1. General Terms of Use and Redistributing Project Gutenberg electronic works

1.A. By reading or using any part of this Project Gutenberg electronic work, you indicate that you have read, understand, agree to and accept all the terms of this license and intellectual property (trademark/copyright) agreement. If you do not agree to abide by all the terms of this agreement, you must cease using and return or destroy all copies of Project Gutenberg electronic works in your possession. If you paid a fee for obtaining a copy of or access to a Project Gutenberg electronic work and you do not agree to be bound by the terms of this agreement, you may obtain a refund from the person or entity to whom you paid the fee as set forth in paragraph 1.E.8.

1.B. “Project Gutenberg” is a registered trademark. It may only be used on or associated in any way with an electronic work by people who agree to be bound by the terms of this agreement. There are a few things that you can do with most Project Gutenberg electronic works even without complying with the full terms of this agreement. See paragraph 1.C below. There are a lot of things you can do with Project Gutenberg electronic works if you follow the terms of this agreement and help preserve free future access to Project Gutenberg electronic works. See paragraph 1.E below.

1.C. The Project Gutenberg Literary Archive Foundation (“the Foundation” or PGLAF), owns a compilation copyright in the collection of Project Gutenberg electronic works. Nearly all the individual works in the collection are in the public domain in the United States. If an individual work is unprotected by copyright law in the United States and you are

located in the United States, we do not claim a right to prevent you from copying, distributing, performing, displaying or creating derivative works based on the work as long as all references to Project Gutenberg are removed. Of course, we hope that you will support the Project Gutenberg mission of promoting free access to electronic works by freely sharing Project Gutenberg works in compliance with the terms of this agreement for keeping the Project Gutenberg name associated with the work. You can easily comply with the terms of this agreement by keeping this work in the same format with its attached full Project Gutenberg License when you share it without charge with others.

1.D. The copyright laws of the place where you are located also govern what you can do with this work. Copyright laws in most countries are in a constant state of change. If you are outside the United States, check the laws of your country in addition to the terms of this agreement before downloading, copying, displaying, performing, distributing or creating derivative works based on this work or any other Project Gutenberg work. The Foundation makes no representations concerning the copyright status of any work in any country other than the United States.

1.E. Unless you have removed all references to Project Gutenberg:

1.E.1. The following sentence, with active links to, or other immediate access to, the full Project Gutenberg License must appear prominently whenever any copy of a Project Gutenberg work (any work on which the phrase “Project Gutenberg” appears, or with which the phrase “Project Gutenberg” is associated) is accessed, displayed, performed, viewed, copied or distributed:

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project GutenbergTM License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

1.E.2. If an individual Project Gutenberg electronic work is derived from texts not protected by U.S. copyright law (does not contain a notice indicating that it is posted with permission of the copyright holder), the work can be copied and distributed to anyone in the United States without paying any fees or charges. If you are redistributing or providing access to a work with the phrase “Project Gutenberg” associated with or appearing on the work, you must comply either with the requirements of paragraphs 1.E.1 through 1.E.7 or obtain permission for the use of the work and the Project Gutenberg trademark as set forth in paragraphs 1.E.8 or 1.E.9.

1.E.3. If an individual Project Gutenberg electronic work is posted with the permission of the copyright holder, your use and distribution must comply with both paragraphs 1.E.1 through 1.E.7 and any additional terms imposed by the copyright holder. Additional terms will be linked to the Project Gutenberg License for all works posted with the permission of the copyright holder found at the beginning of this work.

1.E.4. Do not unlink or detach or remove the full Project Gutenberg License terms from this work, or any files containing a part of this work or any other work associated with Project Gutenberg.

1.E.5. Do not copy, display, perform, distribute or redistribute this electronic work, or any part of this electronic work, without prominently displaying the sentence set forth in paragraph 1.E.1 with active links or immediate access to the full terms of the Project Gutenberg License.

1.E.6. You may convert to and distribute this work in any binary, compressed, marked up, nonproprietary or proprietary form, including any word processing or hypertext form. However, if you provide access to or distribute copies of a Project Gutenberg work in a format other than “Plain Vanilla ASCII” or other format used in the official version posted on the official Project Gutenberg website (www.gutenberg.org), you must, at no additional cost, fee or expense to the user, provide a copy, a means of exporting a copy, or a means of obtaining a copy upon request, of the work in its original “Plain Vanilla ASCII” or other form. Any alternate format must include the full Project Gutenberg License as specified in paragraph 1.E.1.

1.E.7. Do not charge a fee for access to, viewing, displaying, performing, copying or distributing any Project Gutenberg works unless you comply with paragraph 1.E.8 or 1.E.9.

1.E.8. You may charge a reasonable fee for copies of or providing access to or distributing Project Gutenberg electronic works provided that:

- You pay a royalty fee of 20% of the gross profits you derive from the use of Project Gutenberg works calculated using the method you already use to calculate your applicable taxes. The fee is owed to the owner of the Project Gutenberg trademark, but he has agreed to donate royalties under this paragraph to the Project Gutenberg Literary Archive Foundation. Royalty payments must be paid within 60 days following each date on which you prepare (or are legally required to prepare) your periodic tax returns. Royalty payments should be clearly marked as such and sent to the Project Gutenberg Literary Archive Foundation at the address specified in Section 4, "Information about donations to the Project Gutenberg Literary Archive Foundation."
- You provide a full refund of any money paid by a user who notifies you in writing (or by e-mail) within 30 days of receipt that s/he does not agree to the terms of the full Project Gutenberg™ License. You must require such a user to return or destroy all copies of the works possessed in a physical medium and discontinue all use of and all access to other copies of Project Gutenberg™ works.
- You provide, in accordance with paragraph 1.F.3, a full refund of any money paid for a work or a replacement copy, if a defect in the electronic work is discovered and reported to you within 90 days of receipt of the work.
- You comply with all other terms of this agreement for free distribution of Project Gutenberg™ works.

1.E.9. If you wish to charge a fee or distribute a Project Gutenberg™ electronic work or group of works on different terms than are set forth in this agreement, you must obtain permission in writing from the Project Gutenberg Literary Archive Foundation, the manager of the Project

Gutenberg™ trademark. Contact the Foundation as set forth in Section 3 below.

1.F.

1.F.1. Project Gutenberg volunteers and employees expend considerable effort to identify, do copyright research on, transcribe and proofread works not protected by U.S. copyright law in creating the Project Gutenberg™ collection. Despite these efforts, Project Gutenberg™ electronic works, and the medium on which they may be stored, may contain “Defects,” such as, but not limited to, incomplete, inaccurate or corrupt data, transcription errors, a copyright or other intellectual property infringement, a defective or damaged disk or other medium, a computer virus, or computer codes that damage or cannot be read by your equipment.

1.F.2. LIMITED WARRANTY, DISCLAIMER OF DAMAGES - Except for the “Right of Replacement or Refund” described in paragraph 1.F.3, the Project Gutenberg Literary Archive Foundation, the owner of the Project Gutenberg™ trademark, and any other party distributing a Project Gutenberg™ electronic work under this agreement, disclaim all liability to you for damages, costs and expenses, including legal fees. YOU AGREE THAT YOU HAVE NO REMEDIES FOR NEGLIGENCE, STRICT LIABILITY, BREACH OF WARRANTY OR BREACH OF CONTRACT EXCEPT THOSE PROVIDED IN PARAGRAPH 1.F.3. YOU AGREE THAT THE FOUNDATION, THE TRADEMARK OWNER, AND ANY DISTRIBUTOR UNDER THIS AGREEMENT WILL NOT BE LIABLE TO YOU FOR ACTUAL, DIRECT, INDIRECT, CONSEQUENTIAL, PUNITIVE OR INCIDENTAL DAMAGES EVEN IF YOU GIVE NOTICE OF THE POSSIBILITY OF SUCH DAMAGE.

1.F.3. LIMITED RIGHT OF REPLACEMENT OR REFUND - If you discover a defect in this electronic work within 90 days of receiving it, you can receive a refund of the money (if any) you paid for it by sending a written explanation to the person you received the work from. If you received the work on a physical medium, you must return the medium with your written explanation. The person or entity that provided you with the defective work may elect to provide a replacement copy in lieu of a refund. If you received the work electronically, the person or entity providing it to

you may choose to give you a second opportunity to receive the work electronically in lieu of a refund. If the second copy is also defective, you may demand a refund in writing without further opportunities to fix the problem.

1.F.4. Except for the limited right of replacement or refund set forth in paragraph 1.F.3, this work is provided to you 'AS-IS', WITH NO OTHER WARRANTIES OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO WARRANTIES OF MERCHANTABILITY OR FITNESS FOR ANY PURPOSE.

1.F.5. Some states do not allow disclaimers of certain implied warranties or the exclusion or limitation of certain types of damages. If any disclaimer or limitation set forth in this agreement violates the law of the state applicable to this agreement, the agreement shall be interpreted to make the maximum disclaimer or limitation permitted by the applicable state law. The invalidity or unenforceability of any provision of this agreement shall not void the remaining provisions.

1.F.6. INDEMNITY - You agree to indemnify and hold the Foundation, the trademark owner, any agent or employee of the Foundation, anyone providing copies of Project Gutenberg™ electronic works in accordance with this agreement, and any volunteers associated with the production, promotion and distribution of Project Gutenberg™ electronic works, harmless from all liability, costs and expenses, including legal fees, that arise directly or indirectly from any of the following which you do or cause to occur: (a) distribution of this or any Project Gutenberg work, (b) alteration, modification, or additions or deletions to any Project Gutenberg work, and (c) any Defect you cause.

Section 2. Information about the Mission of Project Gutenberg

Project Gutenberg is synonymous with the free distribution of electronic works in formats readable by the widest variety of computers including obsolete, old, middle-aged and new computers. It exists because of the

efforts of hundreds of volunteers and donations from people in all walks of life.

Volunteers and financial support to provide volunteers with the assistance they need are critical to reaching Project Gutenberg's goals and ensuring that the Project Gutenberg collection will remain freely available for generations to come. In 2001, the Project Gutenberg Literary Archive Foundation was created to provide a secure and permanent future for Project Gutenberg and future generations. To learn more about the Project Gutenberg Literary Archive Foundation and how your efforts and donations can help, see Sections 3 and 4 and the Foundation information page at www.gutenberg.org.

Section 3. Information about the Project Gutenberg Literary Archive Foundation

The Project Gutenberg Literary Archive Foundation is a non-profit 501(c)(3) educational corporation organized under the laws of the state of Mississippi and granted tax exempt status by the Internal Revenue Service. The Foundation's EIN or federal tax identification number is 64-6221541. Contributions to the Project Gutenberg Literary Archive Foundation are tax deductible to the full extent permitted by U.S. federal laws and your state's laws.

The Foundation's business office is located at 41 Watchung Plaza #516, Montclair NJ 07042, USA, +1 (862) 621-9288. Email contact links and up to date contact information can be found at the Foundation's website and official page at www.gutenberg.org/contact

Section 4. Information about Donations to the Project Gutenberg Literary Archive Foundation

Project Gutenberg™ depends upon and cannot survive without widespread public support and donations to carry out its mission of increasing the number of public domain and licensed works that can be freely distributed in machine-readable form accessible by the widest array of equipment

including outdated equipment. Many small donations (\$1 to \$5,000) are particularly important to maintaining tax exempt status with the IRS.

The Foundation is committed to complying with the laws regulating charities and charitable donations in all 50 states of the United States. Compliance requirements are not uniform and it takes a considerable effort, much paperwork and many fees to meet and keep up with these requirements. We do not solicit donations in locations where we have not received written confirmation of compliance. To SEND DONATIONS or determine the status of compliance for any particular state visit www.gutenberg.org/donate.

While we cannot and do not solicit contributions from states where we have not met the solicitation requirements, we know of no prohibition against accepting unsolicited donations from donors in such states who approach us with offers to donate.

International donations are gratefully accepted, but we cannot make any statements concerning tax treatment of donations received from outside the United States. U.S. laws alone swamp our small staff.

Please check the Project Gutenberg web pages for current donation methods and addresses. Donations are accepted in a number of other ways including checks, online payments and credit card donations. To donate, please visit: www.gutenberg.org/donate.

Section 5. General Information About Project Gutenberg electronic works

Professor Michael S. Hart was the originator of the Project Gutenberg concept of a library of electronic works that could be freely shared with anyone. For forty years, he produced and distributed Project Gutenberg eBooks with only a loose network of volunteer support.

Project Gutenberg eBooks are often created from several printed editions, all of which are confirmed as not protected by copyright in the U.S. unless a

copyright notice is included. Thus, we do not necessarily keep eBooks in compliance with any particular paper edition.

Most people start at our website which has the main PG search facility:
www.gutenberg.org.

This website includes information about Project Gutenberg, including how to make donations to the Project Gutenberg Literary Archive Foundation, how to help produce our new eBooks, and how to subscribe to our email newsletter to hear about new eBooks.